```
import torch
import torch.nn as nn

class DL_Model(nn.Module):
    def __init__(self):
        super(DL_Model, self).__init__()
 ...
 self.flatten=nn.Flatten()
 self.fc1=nn.Linear(in_features=512*7*7,
                    out_features=1024)
 ...
    def forward(self, x):
        ...
        x=self.flatten(x)
        x=self.fc1(x)
        ...
        return x
```

(a) Draft Code with Bug

```
import torch
import torch.nn as nn

class DL_Model(nn.Module):
    def __init__(self):
        super(DL_Model, self).__init__()
 ...
 self.flatten=nn.Flatten()                    Localized Root Cause
 self.fc1=nn.Linear(in_features=<mask>,
                    out_features=1024)
 ...
    def forward(self, x):
        ...
        x=self.flatten(x)
        x=self.fc1(x)                         Failure Site
        ...
        return x
```

(b) Parameter-Masked Code

```
import torch
import torch.nn as nn

class DL_Model(nn.Module):
    def __init__(self):
        super(DL_Model, self).__init__()
 ...
 self.flatten=nn.Flatten()
 self.fc1=nn.Linear(in_features=<mask>,
                    out_features=1024)
 ...
    def forward(self, x):
        ...                  #tensor shape:(1,512,28,28)
        x=self.flatten(x)  #tensor shape:(1,401408)
        x=self.fc1(x)
        ...
        return x
```

(c) Shape-Annotated Code

```
import torch
import torch.nn as nn

class DL_Model(nn.Module):
    def __init__(self):
        super(DL_Model, self).__init__()
 ...
 self.flatten=nn.Flatten()
 self.fc1=nn.Linear(in_features=401408,
                    out_features=1024)
 ...
    def forward(self, x):
        ...
        x=self.flatten(x)
        x=self.fc1(x)
        ...
        return x
```

(d) Repaired Code